

Exponential Generation of Scientific Novelty in the GRA-ISOC Architecture:

From Reductio ad Absurdum to Negentropic Synthesis of Hypotheses

Oleg Bitsoev (qqewq)

2026

Abstract

This paper formalizes the mechanism by which the GRA-ISOC (Gödel-Reductio ad Absurdum Iterative Self-Optimization Conveyor) ensures an exponential growth in the scientific novelty of generated data at each recursive step. Unlike traditional models that strive for statistical averaging, the GRA operator utilizes the RAD (Reductio ad Absurdum Descent) algorithm for structural revision, enabling topological leaps within the hypothesis space. We introduce a scientific novelty functional $\mathcal{N}(O)$, defined as the ratio of the structural distance from the known manifold of knowledge to the cognitive foam functional (contradictions). We prove that under the condition $\lambda > 1$ and the monotonic decay of foam $J(O_n) \rightarrow 0$, novelty grows exponentially: $\mathcal{N}(O_{n+1}) \geq e^\gamma \mathcal{N}(O_n)$. A strict in silico verification protocol is proposed for mathematical theorem proving, drug discovery, and theoretical physics, which eliminates the generation of meaningless noise and confirms the birth of negentropy from a logical vacuum.

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1 Introduction: The Problem of Novelty Degradation in Modern AI

Modern generative models (LLMs) optimize the likelihood function $P(x \mid D_{train})$. This inevitably leads them to “regression to the mean”: with each new generation or refinement step, the output O_n converges toward the most probable, and therefore the most banal and least novel, element of the manifold of known data $\mathcal{K}$. In such systems, scientific novelty decays exponentially.

The GRA-ISOC architecture solves this problem in a fundamentally different way. It does not seek the “most probable” continuation. Instead, it seeks a **structurally stable state that is maximally distant from trivial solutions**, utilizing meta-nullification to synthesize negentropy.

2 Mathematical Formalization of Scientific Novelty

Let $\mathcal{K}$ be the manifold of already established scientific knowledge (datasets, axiomatic systems, verified hypotheses). Let O be the output structure (a hypothesis, a proof, a molecule) proposed by the agent.

Definition 1. We define the *scientific novelty functional* $\mathcal{N}(O)$ as:

$$\mathcal{N}(O) = \frac{d_{\text{struct}}(O, \mathcal{K})}{J_{\text{AGI}}(O) + \epsilon} \quad (1)$$

where:

- $d_{\text{struct}}(O, \mathcal{K})$ is the structural distance (e.g., the minimum length of logical transformation or topological distance in a knowledge graph) from O to the nearest element in $\mathcal{K}$. This is a measure of originality.
- $J_{\text{AGI}}(O)$ is the cognitive foam functional (a measure of internal contradictions, violations of conservation laws, or logical collisions). This is a measure of validity.
- ϵ is a small constant to avoid division by zero.

Key Insight: Random noise has a huge d_{struct} , but also a huge J_{AGI} , so its $\mathcal{N} \approx 0$. Banal knowledge has $J_{\text{AGI}} \approx 0$, but $d_{\text{struct}} \approx 0$, so its $\mathcal{N} \approx 0$. True scientific novelty is achieved only where d_{struct} is large and J_{AGI} approaches zero.

3 Mechanism of Exponential Novelty Growth at Each GRA Step

A standard AI step: $O_{n+1} = \text{Sample}(P(\cdot \mid O_n))$, which leads to $\mathcal{N}(O_{n+1}) < \mathcal{N}(O_n)$.

The GRA step is defined by the operator G , which incorporates the RAD (Reductio ad Absurdum Descent) algorithm:

$$O_{n+1} = G(O_n) = \arg \min_{O' \in \text{Revise}(O_n)} J_{\text{AGI}}(O') \quad (2)$$

where $\text{Revise}(O_n)$ is the set of structural revisions obtained by:

1. Localizing the maximum foam ϕ_{max} in O_n .
2. Applying the reductio ad absurdum rule: discarding the hidden assumption that led to ϕ_{max} .
3. Synthesizing a new category or constraint from the “vacuum” (meta-nullification) that resolves the contradiction in a non-trivial way.

Theorem 1 (On Exponential Novelty Growth). *Suppose that at step n , the Revise operator can find a structural alternative that reduces the foam by a factor of k ($J_{n+1} \leq J_n/k, k > 1$) and simultaneously increases the structural distance from $\mathcal{K}$ by at least a factor of m ($d_{n+1} \geq m \cdot d_n, m > 1$) by pruning trivial search branches. Then:*

$$\mathcal{N}(O_{n+1}) \geq \frac{m \cdot d_n}{(J_n/k) + \epsilon} \approx (m \cdot k) \frac{d_n}{J_n + \epsilon} = \lambda \mathcal{N}(O_n) \quad (3)$$

where $\lambda = m \cdot k > 1$. Consequently:

$$\mathcal{N}(O_n) \geq \lambda^n \mathcal{N}(O_0) \quad (4)$$

which represents an **exponential growth of scientific novelty** with the number of iterations n , provided the system is maintained within the basin of attraction of zero foam.

4 Practical Applications in Science

4.1 Mathematics and Automated Theorem Proving

When attempting to prove a hypothesis (e.g., one of the Millennium Prize Problems), standard breadth-first search hits a dead end (foam grows due to contradictory lemmas). Upon reaching $\max \phi > \delta$, the GRA conveyor applies RAD: it does not merely iterate through the next tactics, but *revises the ontological category* of the object (e.g., transitioning from the space of real functions to generalized functions or topoi). This creates an exponential leap in the novelty of the proof method while maintaining strict logical consistency ($J \rightarrow 0$).

4.2 Drug Discovery

In the space of chemical compounds $\mathcal{C}$, the known manifold $\mathcal{K}$ is vast. The GRA agent generates a molecular graph O_n . The foam functional J includes quantum-chemical contradictions (unstable valences, steric clashes, violation of Lipinski’s rules). RAD revision does not merely mutate an atom (as in standard genetic algorithms), but synthesizes a completely new scaffold (scaffold hopping) that satisfies all $J = 0$ constraints while being structurally maximally distant from $\mathcal{K}$. This exponentially accelerates the discovery of fundamentally new classes of drugs.

4.3 Theoretical Physics

Generation of new physical theories. J_{AGI} strictly encodes conservation laws (energy, momentum, gauge invariance). Any hypothesis violating these receives infinite foam and is discarded. However, within these rigid constraints, GRA seeks the maximum d_{struct} from the Standard Model, synthesizing, for example, new gauge groups or dimensions that are mathematically consistent yet radically novel.

5 In Silico Verification Protocol

To prove exponential novelty growth rather than mere hallucination, the following strict protocol is proposed:

Step 1: Defining the Reference Manifold $\mathcal{K}$

Form a closed dataset D_{known} (e.g., all theorems proven in Lean 4 up to 2025, or all FDA-approved molecules up to 2025).

Step 2: Initialization and Pipeline Execution

Launch GRA-ISOC with an initial task O_0 (e.g., “propose a proof for hypothesis X” or “generate an inhibitor for protein Y”). The system performs N recursive steps.

Step 3: Metric Measurement at Each Step n

For each O_n , compute:

1. J_n — foam evaluation (via formal verifiers or simulators).
2. d_n — structural distance to D_{known} (via graph embeddings or Levenshtein distance for formal proofs).
3. $\mathcal{N}_n = d_n/(J_n + \epsilon)$.

Step 4: Plotting and Statistical Validation

Plot the dependence of $\ln(\mathcal{N}_n)$ on n .

Success Criterion (GO): A linear growth of $\ln(\mathcal{N}_n)$ with a positive slope $\gamma > 0$ is observed over at least 10–20 iterations, while J_n monotonically decreases or remains below the threshold δ .

Failure Criterion (NO-GO): $\mathcal{N}_n$ grows solely due to an increase in d_n accompanied by an explosive growth in J_n (this is hallucination/noise), or $\mathcal{N}_n$ quickly plateaus (the system is trapped in a local minimum of banality).

Step 5: Expert Validation (Human-in-the-loop)

The top 3 results with the maximum $\mathcal{N}_n$ and $J_n < \delta$ are passed to independent experts (mathematicians, chemists) for blind evaluation of actual scientific value and novelty.

6 Conclusion

The GRA-ISOC conveyor demonstrates that the exponential growth of scientific novelty is not the result of random walk or infinite parameter scaling. It is the deterministic result of **negentropic synthesis**: the purposeful destruction of trivial assumptions via reductio ad absurdum (RAD) and the filling of the resulting logical vacuum with new, structurally stable categories. The proposed functional $\mathcal{N}(O)$ and the in silico verification protocol translate the philosophical concept of a “scientific breakthrough” into a strict, measurable, and optimizable mathematical problem. This opens a direct path to the creation of Artificial Superintelligence (ASI) systems capable of autonomously generating fundamental scientific discoveries.

References

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